
DATA SCIENCE RESEARCH PROJECT

Explainable Peer-Group Anomaly Detection in Individual Tax Returns

A Segment-Conditional Isolation Forest Framework Applied to Australian Taxation Office
Individual Sample Data 2022–2023

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By

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Abstract

Tax authorities must identify from millions of self assessed tax returns, the small number of reports who have anomaly. Confirmed non-compliance labels are rare and selectively observed based on auditors point of view, hence using machine learning to identify those anomaly in taxpayers would save time and effort. Unsupervised anomaly detection could be a solution to this problem, but scoring every taxpayer against the entire population could bring two issues. One is that returns that are irregular, and returns that are rare kind, such as landlords, sole traders or self-funded retirees. This paper proposes and implements a four-stage framework that segments taxpayers into peer groups using K-Means method, then detects anomalous reporting within each group using Isolation Forest, then tests how far anomaly mechanisms can be learned by a supervised Random Forest method, and finally explains both segment and anomaly using SHAP. K-Means selected eight well separated segments with silhouette 0.65 that correspond to recognisable taxpayer types. And for every segment, we run Isolation Forest on 56 behavioural and amount features. Then compared it with a population-wide forest scores. Random Forest and a gradient-boosting has been trained on variables created by injecting five documented anomaly mechanisms. SHAP explains the segmentation through a surrogate model and the forests through exact TreeExplainer attributions. Segmentation removed the population-wide forest's overflagging of rare taxpayer types, but it did not raise recall of the injected anomalies. The Random Forest model recovered almost all injected records with F1 score 0.917 and PR-AUC 0.975. Training to supervised model was done using injected anomalies, not to real non-compliance records. Flagged return in this research is rather a statistical anomaly, not a compliance-risk indicator.

Keywords: tax anomaly detection; taxpayer classification; taxpayer segmentation; unsupervised machine learning; supervised machine learning; Isolation Forest; clustering; K-Means; explainable AI; SHAP; tax compliance; anomaly detection; tax risk profiling.

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1 Introduction

1.1 Background and Research Context

Australian taxation system is based on self-assessment which means they accept the majority of returns as submitted by individuals and depends on data verification which is collected afterward from different sources. The ATO audit results show that people claim high deductions, not reporting their income, provide false information to declare their tax return. The misclaimed deductions like work-related expenses and rental deductions account for shortfall among individuals not operating businesses. Most common issues include that claims have been submitted without any receipts, costs that is only for personal use, and etc. For example, the common things are that someone might claim the full cost of a mobile phone that is also used for personal reasons [1]. Tax administrations are requested to improve their efficiency and accountability while reducing budgets and staff requirements.

ATO estimates tax gap each year which is the difference between what the ATO expects to collect and the amount that would have been collected if every taxpayer was fully compliant with relevant taxation law. The individuals net tax gap is the second largest component in tax gap, at 21% of the overall gap. ATO estimate the tax gap for individuals is \$12.5 billion in 2022–23, or 6.2% of the total theoretical tax, down from 6.7% in 2021–22 [2].

Tax administrations operate under a structural constraint. The taxpayer population is very large, the number of returns that can be examined in detail is small, and examining a compliant taxpayer imposes a cost on both the administration and the taxpayer. In Australia more than sixteen million individual income tax returns are lodged each year [1], and only a small fraction can receive individual attention. As such, deciding which tax return to review or audit is the most time consuming and analytical job with high cost.

1.2 Research Problem and Motivation

The detection of tax anomaly or risk can be effectively approached by methods based on supervised machine learning techniques. However, those methods require large training datasets containing data instances corresponding to both verified tax evasion cases and regular (compliant) tax behavior. As indicated in many previous studies reviewed in the related work section, such datasets are usually not available in many cases. As it contains many sensitive information such as personal details including TFN numbers, financial information and etc, these data are usually protected under strict regulations, making it inaccessible for individual researchers. The second problem with supervised machine learning is that a small number of frauds identified by tax audits that are recorded in the training dataset. Even though having those audit records does not mean those cases are representative of the whole population, as such the trained model will be biased, meaning it will have a high precision but low recall.

Due to limited access to labeled data that required for supervised learning, the detection of tax anomaly or risk is usually approached by unsupervised machine learning methods based on anomaly detection algorithms. Compared with supervised approaches, unsupervised methods

are less precise: they will not identify tax fraud cases, but will point to entities with irregular and suspicious tax behavior including also dishonest tax payers. Therefore, unsupervised methods are suitable as a decision support tool in tax anomaly or risk management systems enabling better prioritization of tax controls and improving efficiency of tax collection. Second, a tax risk management system based on accurate unsupervised learning may lead to a more efficient use of resources when operationally treating identified risks [3].

While these models have shown promising results in analyzing tax-related behaviors, the majority of research tends to concentrate on the development of models, lacking systematic comparative research on multi-source tax-related data integration, industry variability handling, and model adaptability. In light of increasing complexity of tax-related behaviors, a key challenge is how to create high-performance and interpretable risk prediction models based on multi-dimensional variables. The study combines financial variables and non-financial features, but it does not have time-series data to build a comprehensive tax risk assessment of taxpayer due to data restrictions mentioned earlier. comprehensive analysis framework using SVM, XG-Boost, and Random Forest models, to comprehensively improve the accuracy of tax-related behavior identification and the scientific nature of risk management.

The research finds that a multi feature system significantly outperforms single features, showing outstanding performance in capturing tax-related behavior patterns and dynamic changes, with Isolation Forest performing the best, significantly outperforming other models. In addition, service industry enterprises exhibit more stable model classification performance due to the relatively more uniform distribution of features. This result indicates that the optimization of the multi-indicator feature system and the adaptation of model selection are of great significance for enhancing the scientific nature of intelligent analysis and risk management of tax-related behavior.

The research problem is that population-wide anomaly detection on tax returns conflates taxpayer heterogeneity with irregular reporting, and that unsupervised scores are hard to evaluate and to explain.

1.3 Research Questions and Objectives

The objectives of this research paper is to design, implement and evaluate framework that

- Segments taxpayers into groups which is explainable and distinct
- Detect anomaly score for each taxpayer using Isolation Forest
- Test the anomaly mechanism using injected anomaly score by supervised model
- Explain segment anomaly flags using SHAP

Four questions are addressed:

- **RQ1.** Can unsupervised K-Means segment the taxpayer into peer groups that are statistically well separated and interpretable?

- **RQ2.** Does anomaly scores within peer group change which taxpayer will be flagged compared to population wide anomaly score?
- **RQ3.** How well can supervised machine learning can learn the anomaly score within the dataset?
- **RQ4.** What made does anomaly score flag those taxpayers, can SHAP describe these?

1.4 Significance and Expected Contribution

The key contributions of this paper are listed below.

- Existing solutions of the k-means algorithm along with a tax return dataset are implemented and discussed in order to see whether it would segment the correct peer-group variants and the relationships with income stream.
- This research frames the tax authority problems in identifying the risky taxpayer behaviour, analyses and presents a unsupervised machine learning study using the Isolation Forest algorithm.
- Experiments are performed using the injected anomaly scores to find out their effectiveness.

1.5 Thesis Outline

The remainder of this thesis is organised as follows.

- **Chapter 2 – Literature Review** reviews the existing approaches to tax fraud and anomaly detection, from traditional audit and rule-based systems to supervised and unsupervised machine learning, the anomaly detection algorithms used in this study, and explainable AI methods such as SHAP.
- **Chapter 3 – Research Methodology** describes the four-stage framework, the ATO 2022–23 Individual Sample File, the preprocessing and feature engineering steps, the injection of five anomaly mechanisms, and the K-Means segmentation, Isolation Forest, supervised learning and SHAP explainability stages.
- **Chapter 4 – Analysis, Experiments and Results** presents the results for each research question: the segmentation results (RQ1), the segment-based versus population-wide anomaly detection results (RQ2), the supervised learning results (RQ3) and the SHAP explanations (RQ4).
- **Chapter 5 – Discussion** discusses the limitations of the study, including the absence of genuine compliance labels, false positives, data confidentiality and the single-year scope.
- **Chapter 6 – Conclusion and Recommendations** summarises the findings, states the overall contribution of the research and outlines directions for future work.

2 Literature Review

2.1 Traditional and Data-Mining Approaches to Tax Fraud Detection

According to Castellón, tax authorities have traditionally tackled tax fraud through two approaches: auditors experience and rule-based systems [4]. The former approach consists of randomly selecting tax declarations and auditing them based on the experience, intuition, and domain knowledge of the tax auditors. The latter uses methodologies based on rule-based systems. A rule-based system, as described by Baesens et al., is often implemented in the form of a set of if-then rules that detect fraud cases [5]. These rules are developed by a burdensome process, where auditors, after a thorough review, detect a tax fraud case, generalize its characteristics and include a rule into the tax fraud knowledge base. Once applied to incoming cases it triggers an alert or signal when fraud is detected. However, these traditional techniques have two important disadvantages: (1) They rely mainly on past experiences, therefore are unable to discover new fraud mechanisms by themselves. (2) The subjective judgment of experts makes the knowledge bases for rule-based systems expensive to build, maintain and update [5]. A more recent approach to detect tax fraud is through the use of data mining techniques, which provide mechanisms to extract and generate knowledge from substantial volumes of data to support the detection of fraudulent behavior, improving the use of resources [4]. Also, this is one of the best established applications of data mining in both government and industry. Tax authorities in most countries have frequently adopted data mining techniques to assist them in identifying taxpayers who evade obligations [4]. There are works, in the literature, using supervised learning techniques because of the use of labeled data or audit-assisted data.

2.2 Machine Learning for Tax Risk and Financial Anomaly Detection

The introduction of machine learning technology has transformed tasks such as anomaly detection in corporate financial data, tax risk prediction, and compliance auditing from static analysis to dynamic modeling [6, 7]. Research based on different models and algorithmic frameworks has gradually expanded, building efficient risk early warning systems and methods for detecting tax-related behavior, which not only provide strong support for financial supervision but also offer intelligent pathways for enterprises to optimize tax compliance strategies [8].

Financial anomaly detection is an important application of machine learning technology. Researchers have built efficient anomaly identification models to analyze and warn of potential abnormal behavior in corporate financial data. Wang et al. developed an XGBoost ensemble model to identify abnormal tax burden levels among enterprises, significantly enhancing classification performance under imbalanced sample conditions [9]. Mojahedi et al. compared the performance of Logistic Regression, Decision Tree, and Support Vector Machine (SVM) on tax-related datasets, finding that Logistic Regression performed stably when dealing with

low-dimensional data, while SVM was more suitable for anomaly detection in nonlinear scenarios [10]. Zhang et al. designed a triple feature selection mechanism based on the Random Forest model, which effectively improved detection accuracy and stability under multi-dimensional data conditions [11]. Liu et al. further proposed a distribution-transformation-based Random Forest approach, demonstrating strong performance in detecting abnormal behaviors in enterprises' inter-period financial fluctuations [12]. Meanwhile, George et al., through empirical research, found that traditional machine learning models still hold significant application value in identifying financial fraud risks, particularly in enterprise scenarios with moderately sized data samples and clearly defined feature variables [13].

2.3 Unsupervised Approaches

While most studies addressing tax fraud detection employ supervised learning, the lack of historically labeled data limits such studies in most situations since manual auditing and labeling data is expensive and time-consuming. As a result, they proposed using unsupervised learning [14]. The research of de Roux et al. was divided into three phases, which are clustering similarly valued tax declarations; adjusting probability distribution to each cluster tax base, and detecting suspicious activities by checking the quantiles of cluster adjusted distribution. Despite the fact that their model produced promising results, there was the limitation of the limited amount of data, and a low number of variables needed to deeply describe cases. The most worrying issue was that the model's accuracy would not be evaluated because the data was not labeled, which raises questions about its trustworthiness [15].

2.4 Anomaly Detection Methods

Chandola et al. [16] provide the standard taxonomy; the profiles sought here are point anomalies in a multivariate financial space. Anomaly is defined compared to the chosen model normality [17]. Isolation Forest [18, 19] isolates records by splitting them using randomly selected feature. Those which require fewer splits are considered as anomalous. The Local Outlier Factor [20] instead compares a record's local density with that of its nearest neighbours. Comparative studies find no universally superior detector: Goldstein and Uchida [21] found that relative performance depends on whether anomalies are global or local, Campos et al. [22] documented strong sensitivity to parameter choice, and Zimek et al. [23] analyse neighbourhood-based detectors in high-dimensional spaces where distance contrast deteriorates.

2.5 Explainable Artificial Intelligence

The use of Explainable Artificial Intelligence (XAI) methods, including SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations), is becoming more widespread to explain the predictions of the model, so that they can be regulated and trusted by the auditors [24].

2.6 Research Gap and Synthesis

The literature shows that most tax fraud detection studies rely on supervised learning, which requires labelled audit outcomes that are rarely available, while the unsupervised studies that do exist score every taxpayer against the whole population and offer little evaluation or explanation of their results. Comparative studies also show that no anomaly detector is universally superior and that the reference population and feature space strongly influence what is flagged. This research addresses these gaps by segmenting taxpayers into peer groups before applying Isolation Forest, evaluating the results against injected anomalies and a supervised comparator, and explaining both segment membership and anomaly flags with SHAP, as set out in the research questions in Chapter 1.

3 Research Methodology

3.1 Overall Framework

Fig. 1 summarises the pipeline: raw taxpayer data → preprocessing → feature engineering → Stage 1 segmentation → peer-group formation → Stage 2 Isolation Forest within each peer group → Stage 3 supervised classification → Stage 4 SHAP → risk interpretation and visualisation. Segmentation precedes anomaly detection because the reference population must be fixed before “unusual” can be defined: a forest fitted to the whole population isolates whatever is globally rare, whereas a forest fitted within a peer group isolates what is rare among similar taxpayers. The supervised stage uses injected labels to measure how far documented mechanisms are learnable and to provide a calibrated comparator for the unsupervised scores. The SHAP explainability describes both segment and anomaly score into which feature have mostly contributed to anomaly.

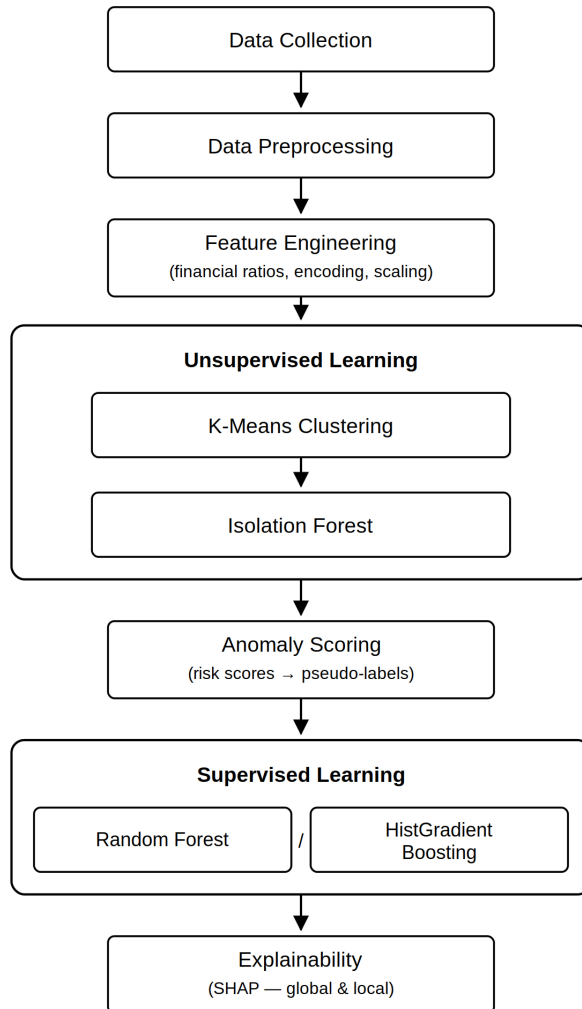


Figure 1: Four-stage taxpayer analytics framework.

3.2 Dataset

The primary data used in this research were obtained from Australian Taxation Authority and it covers the tax filing records of 2% confidentialised random sample for Financial Year 2022 to 2023. It covers around 321,890 anonymised individual income tax returns data released by ATO only for research purposes. The access to the data is limited to the people who have signed Confidentiality Deed with ATO and granted through the ATO’s taxation statistics program. The variable index for these data have sixty-five fields including taxpayer demographics, reported income, claimed deductions, offsets and etc. It has no personal information, such as name, address, DOB, or postcode. These variables group into below summarised groups in Table 1:

Table 1: Dataset and preprocessing summary.

Characteristic	Value
Source	ATO 2022–23 individual sample file (SA4 geography)
Reference population	16,108,843 individual taxpayers
Records in the sample	321,890
Sampling fraction	1.9982%
Variables in the file	65
Missing (blank) cells	0
Fully duplicated records	0
Financial variables available	57

The data has sixty five variables belonging to several groups. Seven of them belongs to taxpayer characteristic group, which have variables of gender, age, occupation, region and etc. Most of the variables fall into income and deduction items, which is shown below. And remaining variables belongs to tax offsets, tax credits, super and etc.

Table 2: Variable groups in the ATO Individuals Sample File and their analytical role.

Variable group	Representative variables	Analytical role
Taxpayer characteristics	Gender, age_range, Occ_code, Partner_status, SA4, Lodgment_method, PHI_Ind	Define peer groups and control for legitimate heterogeneity
Income items	Sw_amt, Alow_ben_amt, Grs_int_amt, Frk_Div_amt, Net_rent_amt, Gross_rent_amt, Net_PP_BI_amt, Net_CG_amt, Tot_IncLoss_amt	Establish the scale and composition of declared income
Deduction items	WRE_car_amt, WRE_trvl_amt, WRE_uniform_amt, WRE_self_amt, WRE_other_amt, Rent_int_ded_amt, Gift_amt, Cost_tax_affairs_amt, Tot_ded_amt	Primary risk detection of over-claiming deduction
Offsets, credits and super	Cr_PAYG_ITI_amt, TFN_amts_wheld_gr_intst_amt, Rptbl_Empr_spr_cont_amt, Spr_Ttl_Acnt_Bal	Cross-checks for a return
Outcome and process fields	Taxable_Income, Hrs_to_prepare_BPI_cnt, Help_debt,	Contextual variables and candidate segmentation cuts

Dataset has no blank cell and no duplicates as these information has been extracted from ATO dataset. Every zero amount reported in the dataset means it was not reported by taxpayer and was treated as information. The ATO sample file does not have any variable for tax amount, tax refund amount, lodgment dates, or audit outcome, so ratios involving tax paid, refunds, late filing cannot be included in this research. As there is no compliance label exist in the data, synthetic anomalies are injected (Section 3.3) in detection metrics.

Fig. 2 Distribution of key financial variables (non-zero values, log scale)

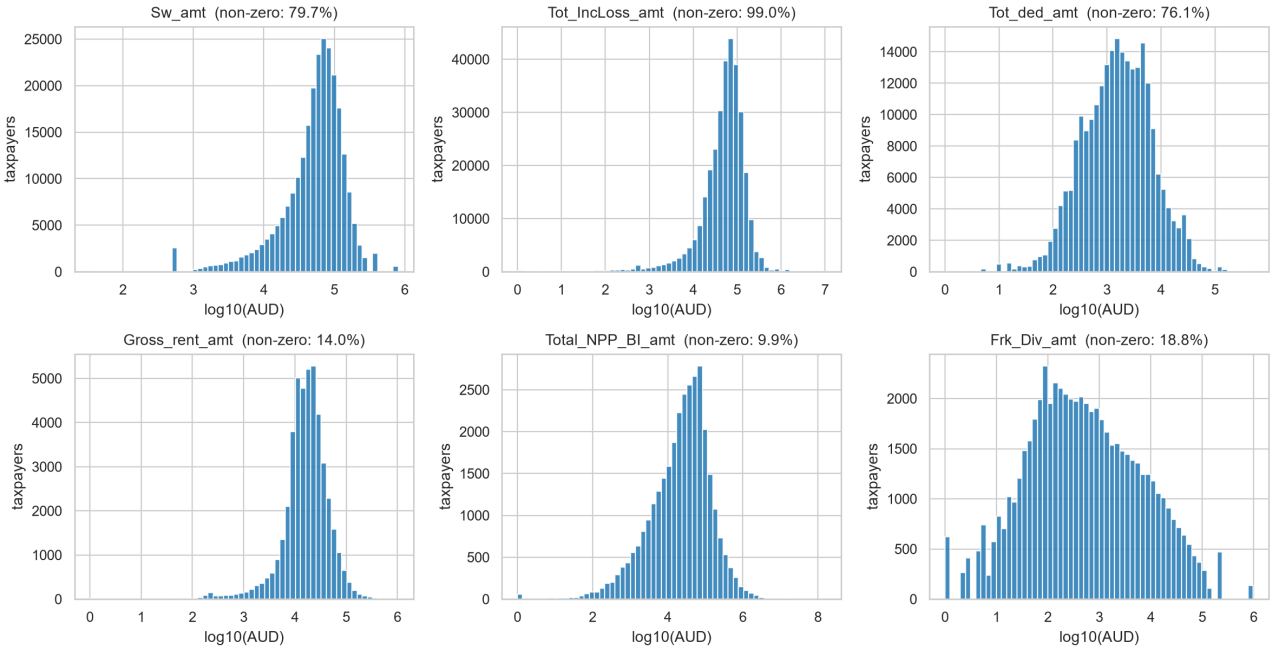


Figure 2: Distribution of key financial variables (non-zero values, log scale).

3.3 Preprocessing and Feature Engineering

Amounts reported by fewer than 1% of taxpayers were removed (ten variables, e.g. primary-production business items), because a variable that is almost always zero would be flagged merely for being non-zero. Of the remainder, one member of each pair with $|r| > 0.95$ on the transformed scale was dropped to avoid double counting: the franking credit ($r = 0.998$ with franked dividends), other rental deductions (0.992 with gross rent) and business expenses (0.951 with business income); their information is retained through ratios. Forty-four amounts remain.

The main dataset included 65 variables including taxpayer characteristics, income items, deductions and etc. In this research, we employed ten financial ratios instead of using the raw variables as relying on raw dollar amounts for anomaly detection could lead to issues where only individuals with significantly high or low incomes, or extreme or low deduction amounts, would be identified as anomalies. Therefore, we utilized these ratios to transform the model’s variables into quantifiable and comparable values that can be assessed among taxpayers without concerns about their income levels and provide a standard format for benchmark assessments. The ratio of work-related expenses to salary is applicable for taxpayers earning a salary, while the rental deduction to rental income ratio serves as a means for anomaly detection among those with rental income, and the ratio of business expenses to business income is relevant for individuals with business income.

The 44 transformed amounts are combined with twelve behavioural features (Table 3). Each ratio divides by $\max(\text{denominator}, \$1,000)$ and is capped, so division by zero cannot occur. The taxable income ratio checks whether the reported taxable income amount matches the calculated amount from other variable including income and deductions.

Table 3: Behavioural features of the research.

Feature	Definition	Description
deduction_to_income_ratio	total deductions / total income	Could signal over-claiming expenses
wre_to_salary_ratio	five work-related expense items / salary	Benchmarked by occupation
business_expense_ratio	business expenses / business income	Could signal over-claiming expenses or unreported income
rental_deduction_ratio	three rental deduction items / gross rent	Rental losses lead to potential audit risk
taxable_income_ratio	taxable income / (income – deductions – losses)	Internally inconsistent return
investment_deduction_ratio	interest and dividend deductions / investment income	Investment deductions without matching income
gift_to_income_ratio	gifts and donations / total income	Very large donation claims compared to income
tax_affairs_cost_ratio	cost of managing tax affairs / total income	Unusually high advisory cost
personal_super_ratio	personal super contributions / total income	Contributions far above income are unusual
payg_instalment_ratio	PAYG instalment credits / total income	Instalments inconsistent with declared income
items_reported	number of non-zero amount fields	Return complexity
lodged_via_agent	1 if lodged by a tax agent	Agent-prepared returns claim differently

As there is no compliance records exists in the dataset, 1% of records of 3,219 tax return details were extracted from dataset to create labelled anomalies, following the benchmark-construction logic of Emmott et al. [25]. Five mechanisms reflecting recognised risk types that the file can represent were applied:

- **A1 Work-related expense inflation** (salary above \$20,000). Work-related expenses raised to 25–45% of salary amount (peer 99th percentile 24%).
- **A2 Business expense inflation** (business income above \$5,000, expenses below income). Expenses raised to 150–300% of business income (peer 99th percentile 220%).
- **A3 Rental deduction inflation** (gross rent above \$5,000). Rental deductions raised to 300–500% of gross rent (peer 99th percentile 295%).
- **A4 Taxable-income identity violation** (income above \$30,000). Taxable income set to 40–70% of the identity-implied amount, nothing else changed.
- **A5 Investment deductions without investment income** (no interest or dividends reported). Interest deductions have been reported for 15–35% of income even though there was no related income.

3.4 Segmentation

The k-means algorithm depends on the value of k and clustering with different k values will eventually produce different results. Candidates for k number were compared using several methods. First we used inertia elbow method, where it measure how internally coherent the clusters are. Second, silhouette method where it shows the separation distance between the resulting clusters [26]. Third, we used Davies–Bouldin Index as it is calculated as the average similarity of each cluster with a cluster most similar to it [27]. Caliński–Harabasz Index is a measure of how similar an object is to its own cluster (cohesion) compared to other clusters (separation) [28]. It has computed on a fixed 20,000 record sample because silhouette is quadratic. One segment have to have at minimum 2,000 records to support its own forest. Among these methods, k number with highest silhouette and lowest Davies–Bouldin Index was selected. Segments were named after their correlated median income.

3.5 Anomaly Detection

One Isolation Forest (200 trees, subsample 256 [18]) was fitted to each segment with 56 features and used to score anomaly of its own clusters. Scores are reported as the negative of scikit-learn’s `score_samples` so that higher means it is more isolated from the segment. Since contamination percentage only cut-off the top anomaly at 1%, 2%, 5% and 10% which taken from segment score quantiles, 2% has been chosen for this research as it is practical rather than an estimate of the unknown true anomaly rate. A population wide forest with identical settings was fitted for comparison. The designs were checked using on overlap score like Jaccard and Spearman, recall and precision of injected anomalies at each budget. A pre-specified test was repeated on both designs on the twelve behavioural ratios alone, to tell the effect of the reference group from the effect of the feature.

3.6 Supervised Learning

Random Forest operates by creating multiple decision trees, each trained on a random subset of the data and features. The injected labels with the same 56 features from Isolation Forest has been used to train Random Forest [29] (300 trees, minimum leaf three, balanced-subsample class weights) and a histogram gradient-boosting classifier [30, 31] (300 iterations, balanced class weights). Data were split into 70 and 30 with stratification. Models were evaluated on the hold out set with accuracy, precision, recall, F1, ROC-AUC, PR-AUC and the confusion matrix at threshold 0.5, and again at the same 2% review budget as the forests so that the paradigms are compared on equal terms.

3.7 SHAP Explainability

SHAP (SHapley Additive exPlanations) calculates a feature’s contribution by considering all possible combinations of features and averaging the marginal contributions across these coali-

tions. A surrogate Random Forest was trained to reproduce the K-Means labels from the nine segmentation features [32]. SHAP evaluates how much each feature contributes to the final prediction across all possible feature combinations. To identify *why was a taxpayer flagged*, TreeExplainer was applied directly to each segment’s Isolation Forest. Tree models make decisions through a series of splits, and SHAP can trace these decision paths to quantify each feature’s contribution with mathematical precision. It uses the structure of tree models to compute exact SHAP values efficiently, making it fast and accurate method for explaining predictions. SHAP values were computed for 300 flagged and 300 unflagged taxpayers per segment for global summaries and individually for selected taxpayers for waterfall explanations. All experiments use seed 42 with scikit-learn 1.9 and SHAP 0.52 [31].

4 Analysis, Experiments and Results

In this section, we will present and discuss the results obtained from the computation.

4.1 Segmentation Results (RQ1)

Silhouette rises from $k = 3$ to a maximum of 0.653 at $k = 8$ and drops to 0.417 at $k = 9$, and the Davies–Bouldin index reaches its minimum of 0.719 at the same k (Table 4, Fig. 3). Every k up to 8 satisfies the minimum segment size, so $k = 8$ was selected for this dataset.

Table 4: Cluster-selection criteria for K-Means (silhouette, Davies–Bouldin and Caliński–Harabasz on a 20,000-record sample).

k	Silhouette	Davies–Bouldin	Caliński–Harabasz	Smallest segment	Largest segment (%)
3	0.550	1.766	4,405.158	22,566	73.300
4	0.576	1.429	4,791.651	14,862	73.400
5	0.605	1.323	5,443.486	14,716	73.100
6	0.621	1.132	6,300.054	5,469	72.800
7	0.643	0.865	7,873.797	5,429	72.600
8	0.653	0.719	8,404.810	4,223	72.500
9	0.417	0.804	8,626.968	4,205	58.200
10	0.421	0.844	8,479.447	3,228	58.000

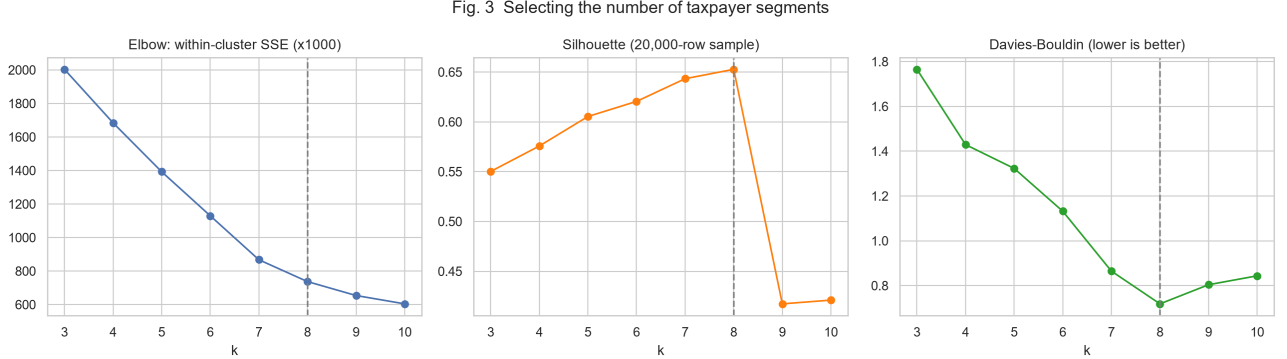


Figure 3: Selecting the number of taxpayer segments.

The eight segments (Table 5, Fig. 4) correspond to recognisable taxpayer types without being imposed. 72.5% of taxpayers with median income of \$67,797 belong to segment S0 and their modal age is between 30 to 34. 5.5% of records are in Segment S1, and it is pension-dominated by 79% population, with modal age is 70 and over and a median super balance of only \$1,722, retiree profile. Segment S2 is 88% of business profile dominant group, which agent-lodged in 90% of cases, median income of \$36,148. Segment S3 is 88% investment dominant with low taxable income but a median super balance of \$141,883. Segment S5, 2.9% of records, is rental dominant, Segment S6 4.5% of records is trust and partnership dominant with 96.5% agent lodgment. Majority of segment S7 reported no amount in income and deuctions with median income of \$0 and only one item being reported.

Table 5: Segment profiles (medians in AUD).

Segment	<i>n</i>	Share (%)	Median total income	Median deductions	Median super balance	Agent lodged (%)	Modal age range	Top occupation
S0: Wage-dominant (mid income)	233,317	72.5	67,797	1,367	57,133	58.4	8	Professionals
S1: Pension/retirement (low income)	17,834	5.5	23,327	0.0	1,721.5	44.5	0	Not stated
S2: Business-dominant (mid income)	18,650	5.8	36,148	0.0	19,855.5	89.6	8	Not stated
S3: Investment-income (low income)	18,758	5.8	16,659	0.0	141,883	76.9	0	Not stated
S4: Other-income (mid income)	5,402	1.7	40,794.5	235.0	40,881.5	69.3	0	Not stated
S5: Rental-property (low income)	9,266	2.9	17,606.5	268.5	72,392.5	86.2	0	Not stated
S6: Trust/partnership (mid income)	14,440	4.5	65,278	184.0	75,413	96.5	0	Not stated
S7: No/minimal reported income	4,223	1.3	0.0	0.0	0.0	36.3	8	Not stated

Fig. 4 Taxpayer segments

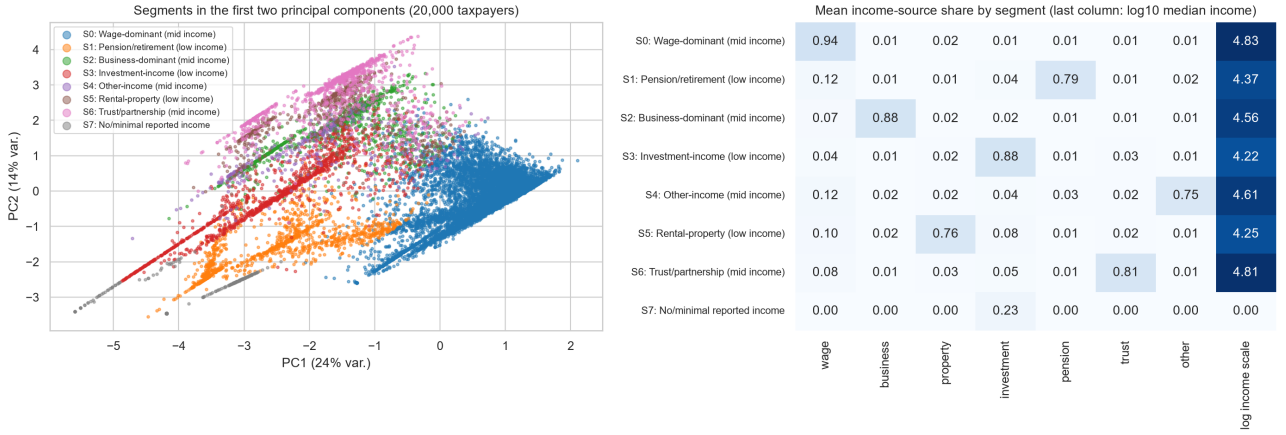


Figure 4: Taxpayer segments in the first two principal components, and mean income-source share by segment.

RQ1 is answered positively that a income composition space yields well-separated and tax-interpretable peer groups, in contrast to the weak separation (silhouette 0.05–0.13) obtained on this same data when only dollar amounts were clustered not ratio.

4.2 Unsupervised Anomaly Detection (RQ2)

At the 2% threshold, the design using segmentation flagged 6,442 records and the population-wide design has 6,438 records respectively. Of the segmented flags, 6,064 (94%) were natural profiles and 378 injected records.

Table 6: Within-segment anomaly-score thresholds and number flagged at each contamination level.

Segment	<i>n</i>	Threshold 1%	Threshold 2%	Threshold 5%	Threshold 10%	Flagged at 2%
S0: Wage-dominant (mid income)	233,317	0.5616	0.5332	0.4915	0.4567	4,667
S1: Pension/retirement (low income)	17,834	0.5603	0.5294	0.4835	0.4472	357
S2: Business-dominant (mid income)	18,650	0.5768	0.5425	0.4960	0.4553	373
S3: Investment-income (low income)	18,758	0.5628	0.5449	0.5102	0.4781	376
S4: Other-income (mid income)	5,402	0.5470	0.5286	0.4921	0.4641	109
S5: Rental-property (low income)	9,266	0.5509	0.5263	0.4929	0.4615	186
S6: Trust/partnership (mid income)	14,440	0.5459	0.5234	0.4892	0.4580	289
S7: No/minimal reported income	4,223	0.5761	0.5394	0.4854	0.4410	85

Fig. 5 Isolation Forest anomaly scores

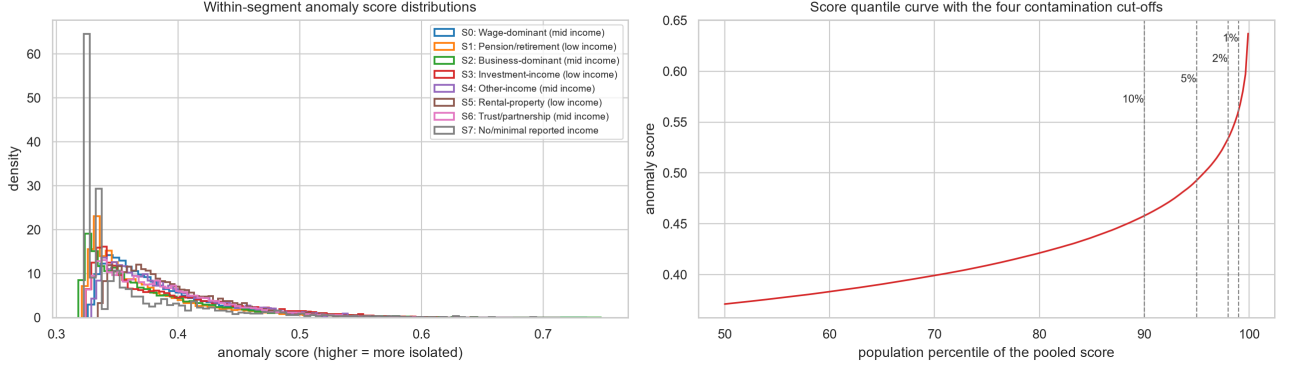


Figure 5: Isolation Forest anomaly scores within segment distribution and the pooled quantile curve with the cut-offs.

Raising the threshold from 1% to 10% raised recall of injected anomalies from 6.8% to 44.7% in segmented anomaly and from 8.3% to 48.8% in population-wide anomaly, while precision against the injected labels fell from about 7% to 4.5% as shown below (Table 7).

Table 7: Contamination results per segment and population.

Contamination	Design	Flagged	Injected recall	Injected precision	Injected F1	Natural flags
0.010	segmented	3,224	0.068	0.068	0.068	3,006
0.010	population-wide	3,219	0.083	0.083	0.083	2,953
0.020	segmented	6,442	0.117	0.059	0.078	6,064
0.020	population-wide	6,438	0.141	0.070	0.094	5,985
0.050	segmented	16,098	0.251	0.050	0.084	15,289
0.050	population-wide	16,095	0.277	0.055	0.092	15,205
0.100	segmented	32,192	0.447	0.045	0.081	30,754
0.100	population-wide	32,189	0.488	0.049	0.089	30,619

The designs agree on 3,909 taxpayers with Jaccard score 0.436 and Spearman correlation of scores 0.795, so it means segmentation changed more than half of the list. The population-wide anomaly concentrated in the small non-wage segments where 8.2% of investment-income taxpayers, 6.2% of landlords, 4.8% of other-income and 4.3% of trust-distribution taxpayers (Fig. 6). These taxpayers were flagged because their type of return compared with all population and majority of population has wage related income. Whereas segmented design equalises this burden at 2% of every peer group. On the pre-specified benchmark, however, segmentation did not improve sensitivity: recall was slightly lower (11.7% versus 14.1%), and by type the segmented forest was better only for work-related-expense inflation (9.6% versus 6.2%), while the population-wide forest recovered more business-expense (21.4% versus 15.2%) and rental-deduction anomalies (31.5% versus 23.1%). The taxable-income identity violation was nearly invisible to both (1.4% and 2.0%) because it perturbs one feature among 56, and isolation rewards records that are extreme on many. Income confounding was not reduced either: 46.9% of segmented and 46.1% of population-wide flags fell in the top income decile, and the score-income rank correlation rose from 0.14 to 0.25, because within the large wage segment scale remains the dominant source of rarity (Table 8).

Table 8: Overlap and income confounding of the two designs at the 2% budget.

	Segmented	Population-wide
Flagged	6,442	6,438
Median income of flagged (AUD)	139,872	135,241.5
Income multiple of population median	2.408	2.328
Flagged in top income decile (%)	46.880	46.070
Flagged below median income (%)	17.665	23.035
Spearman correlation, score vs. income	0.250	0.140

Fig. 6 Segment-based versus population-wide Isolation Forest

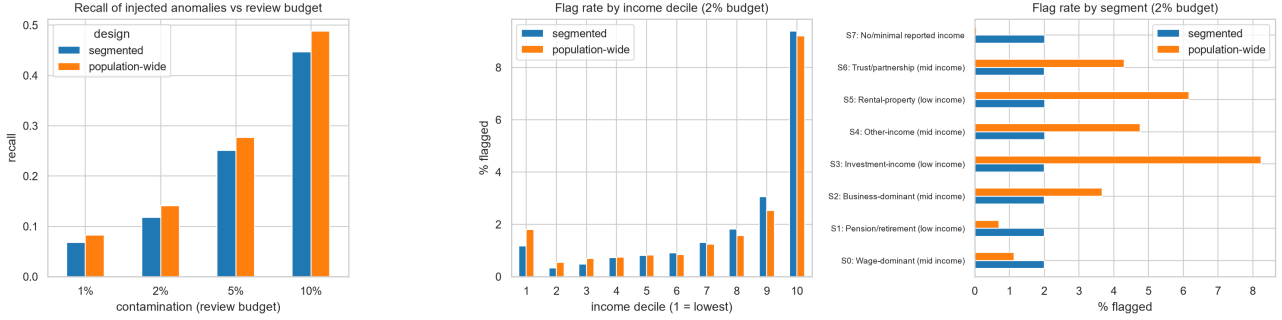


Figure 6: Segment-based versus population-wide Isolation Forest: recall of injected anomalies by budget, flag rate by income decile and by segment.

Repeating both designs on the twelve behavioural ratios alone changed the picture completely (Tables 9 and 10). At the same 2% budget recall of injected anomalies rose from 11.7% to 39.6% (segmented) and from 14.1% to 41.7% (population-wide), precision from 6–7% to 20–21%, and PR-AUC against the injected labels from 0.04–0.05 to 0.22. The share of flags in the top income decile fell from 47% to 15–16%, and the score–income correlation fell to 0.12 (segmented) and to zero (population-wide). Anomaly score for taxpayers with investment deductions without having investment income rose from 9% to 98%, and business expense and rental deduction inflation to 38–49%. Work related expense inflation remained because a 25–45% claim is not extreme relative to the whole ratio distribution, and the identity violation remained undetected. Only 2,370 records were flagged by both the primary and the ratio only methods, so the choice of feature changes the anomaly list as much as the choice of reference population.

Table 9: Ablation: design $\times$ feature space at the 2% budget.

Feature space	Design	Flagged	Injected recall	Injected precision	PR-AUC vs. injected	Flagged in top income decile (%)	Spearman score vs. income
amounts + ratios (primary)	segmented	6,442	0.117	0.059	0.044	46.880	0.250
amounts + ratios (primary)	population-wide	6,438	0.141	0.070	0.051	46.070	0.140
ratios only	segmented	6,442	0.396	0.198	0.216	15.554	0.119
ratios only	population-wide	6,438	0.417	0.208	0.225	14.585	−0.007

Table 10: Recall of each injected mechanism by feature space and design (2% budget).

Anomaly type	amounts + ratios (primary)		ratios only	
	segmented	population-wide	segmented	population-wide
A1 WRE inflation	0.096	0.062	0.214	0.115
A2 Business expense	0.152	0.214	0.376	0.486
A3 Rental deduction	0.231	0.315	0.390	0.470
A4 Taxable identity	0.014	0.020	0.026	0.023
A5 Investment deduction, no income	0.093	0.092	0.975	0.991

The answer to RQ2 is therefore confirmed. Segmentation changes which taxpayers is being reviewed and removes the over representation of rare taxpayer types. Segmentation alone did not make Isolation Forest sensitive at identifying unusual reporting behaviour as the model remained strongly influenced by raw dollar amounts as well. Ratio-based features were needed to remove the effect of scale. Once these behavioural ratios were included, segmentation produced only a small further improvement in anomaly sensitivity.

4.3 Supervised Learning Results (RQ3)

On the set of 96,567 taxpayers (966 injected) the Random Forest achieved precision for 89%, recall for 94% at threshold 0.5, with 109 false positives and 56 false negatives (Table 11, Fig. 7). Gradient boosting have precision at 72% and recall at 98% and trained sixteen times faster. Five-fold cross-validation confirmed the ranking (F1 0.920 ± 0.009 versus 0.856 ± 0.028 ; PR-AUC 0.970 versus 0.974). Accuracy was 99.8% for both.

Table 11: Supervised classification of injected anomalies on the hold-out set.

Model	Threshold	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	TP	FP	FN	TN
Random Forest	0.50	0.998	0.893	0.942	0.917	1.000	0.975	910	109	56	95,492
Random Forest	top 2% budget	0.990	0.499	0.998	0.665	1.000	0.975	964	967	2	94,634
Hist. Gradient Boosting	0.50	0.996	0.727	0.980	0.835	0.997	0.968	947	356	19	95,245
Hist. Gradient Boosting	top 2% budget	0.990	0.496	0.992	0.661	0.997	0.968	958	973	8	94,628

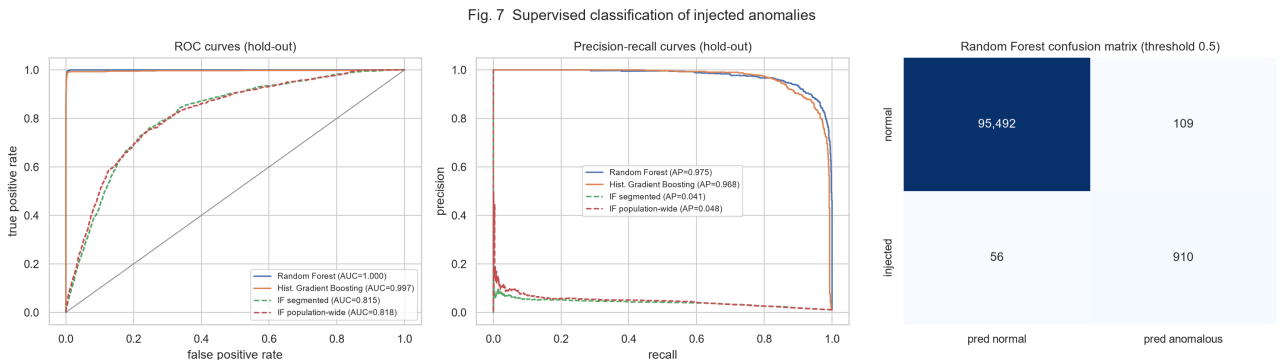


Figure 7: Supervised classification: ROC and precision–recall curves for both classifiers and both forests, and the Random Forest confusion matrix.

Compared on identical hold-out rows and the same 2% budget (Table 12) the supervised models recovered almost every injected record (recall 0.998 and 0.992, precision 0.50) whereas

the two primary forests recovered 10.7% and 12.7% (PR-AUC 0.04–0.05). This gap must not be read as the supervised model being a better fraud detector: it was shown labelled examples of five mechanisms and learned them, while the forests ranked general unusualness with no labels. The two paradigms answer different questions, which their disagreement on natural profiles makes explicit: of the 1,820 unmodified taxpayers flagged by the segmented forest in the hold-out set, the Random Forest labelled 0.6% as anomalous. Operationally, a supervised model generalises only to mechanisms represented in its labels; an unsupervised model retains the ability to surface patterns nobody has labelled.

Table 12: Supervised versus unsupervised detection of injected anomalies on identical hold-out rows and review budget.

Method	Flagged	Precision	Recall	F1	ROC-AUC	PR-AUC
Isolation Forest, segmented (2%)	1,923	0.054	0.107	0.071	0.815	0.041
Isolation Forest, population-wide (2%)	1,926	0.064	0.127	0.085	0.818	0.048
Random Forest (top 2% budget)	1,931	0.499	0.998	0.665	1.000	0.975
Hist. Gradient Boosting (top 2% budget)	1,931	0.496	0.992	0.661	0.997	0.968

4.4 SHAP Results (RQ4)

The Random Forest reproduced the K-Means labels with 99.7% hold-out accuracy, so its explanations can be read as explanations of the segmentation. Wage share was the most influential feature with mean SHAP score of 0.064, followed by business, investment, pension and trust shares (Fig. 8). Each segment is explained mainly by its own dominant share, but the no-income segment is the only one explained chiefly by scale. A taxpayer is placed in a segment because of what kind of income they report, not how much.

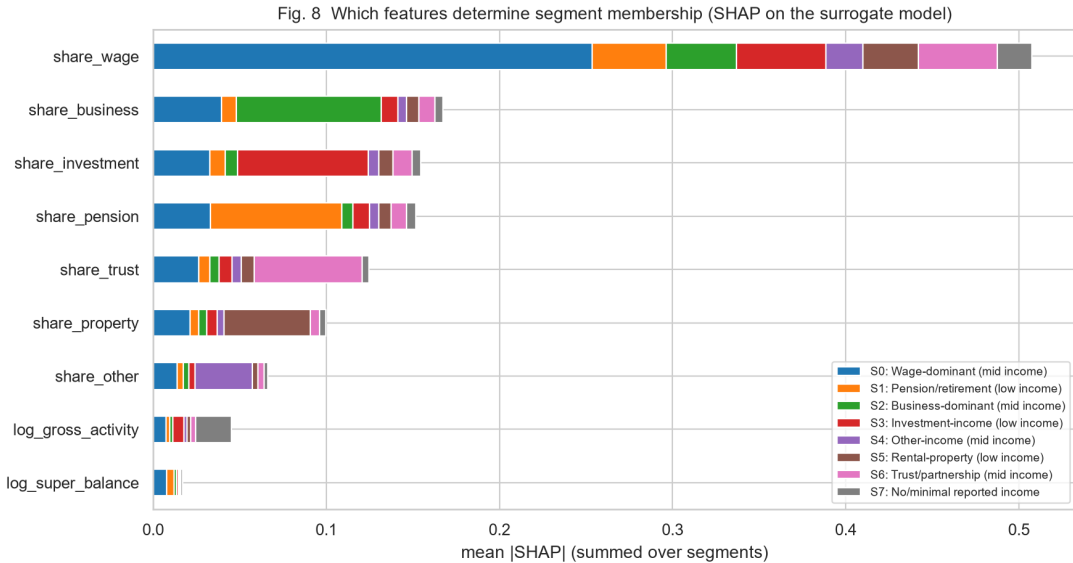


Figure 8: SHAP on the surrogate model: which features determine segment membership, stacked by segment.

The validity check passed in every segment: the sum of SHAP values tracked the forest’s own

score with Spearman correlation 1.000. Pooled over the segment forests (Fig. 9, Table 13) the most influential features were gross rent (mean $|\text{SHAP}|$ 0.196), other foreign income (0.189), total capital gains (0.189), personal superannuation contributions (0.183), gifts (0.166) and net capital gains (0.165); the deduction-to-income ratio ranks thirteenth (0.122) but has the strongest monotone relationship between value and contribution (Spearman 0.92). Every top feature contributes in the same direction: high values push toward “anomalous”, low values are neutral. The primary forests therefore flag taxpayers who report unusual amounts of items most peers do not report at all, and secondarily taxpayers whose deductions are large relative to income; this is the mechanism behind the income confounding and the ablation result above.

Table 13: Global SHAP importance of the segment Isolation Forests (pooled, top 15).

Feature	Mean $ \text{SHAP} $	Mean contribution (flagged)	Mean contribution (unflagged)	Corr. value vs. contribution
Gross_rent_amt	0.196	0.281	−0.002	0.732
Other_foreign_inc_amt	0.189	0.270	−0.007	0.765
Tot_CY_CG_amt	0.189	0.291	−0.002	0.807
Spr_Prsnl_Contr	0.182	0.202	0.005	0.783
Gift_amt	0.166	0.179	0.007	0.827
Net_CG_amt	0.165	0.238	−0.003	0.758
Asbl_forgn_source_incm_amt	0.155	0.216	−0.000	0.787
Unfranked_Div_amt	0.153	0.182	−0.003	0.756
Rent_int_ded_amt	0.141	0.195	−0.005	0.749
Spr_Emplr_Contr	0.131	0.112	−0.006	0.733
Div_Ded_amt	0.124	0.186	−0.003	0.566
Non_emp_spr_amt	0.123	0.166	−0.006	0.685
deduction_to_income_ratio	0.122	0.202	0.002	0.919
Rent_cap_wks_amt	0.122	0.183	−0.000	0.741
WRE_other_amt	0.119	0.127	−0.004	0.638

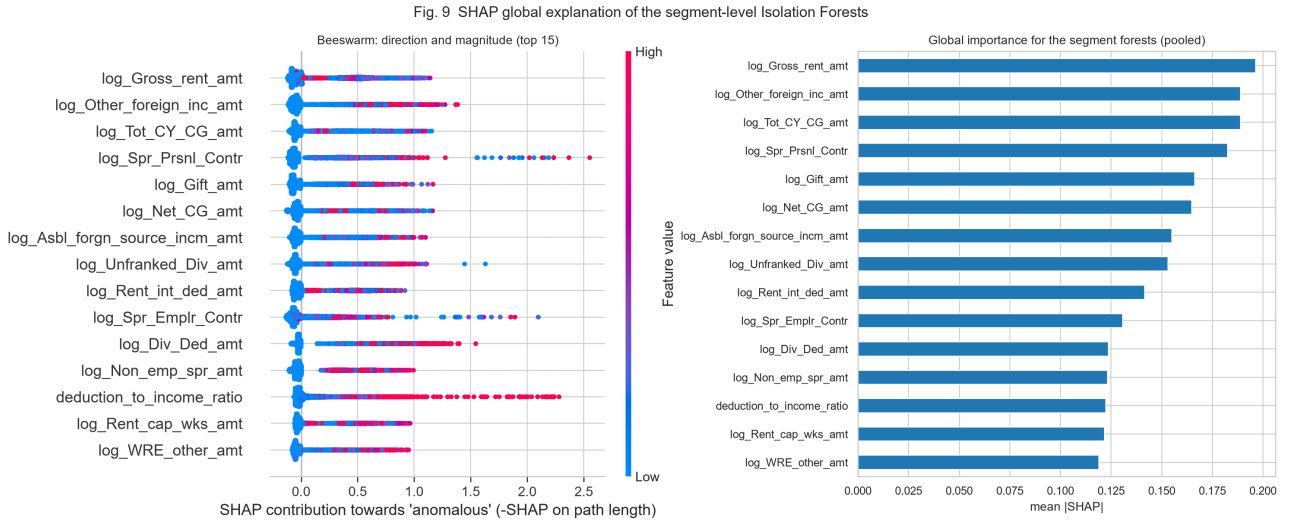


Figure 9: SHAP global explanation of the segment-level Isolation Forests.

No single feature reaches a 50% flag rate: even among the top 1% of taxpayers on gross rent, capital gains or personal superannuation contributions only 15–27% are flagged (Table 14). For rarely reported items the value at which the flag rate first reaches 10% is simply the point at which the item becomes non-zero (other foreign income, unfranked dividends and capital gains: \$1–\$3), and a 25% flag rate is reached only for other foreign income above \$380 and rental interest deductions above about \$59,000. The quantities that actually determine a flag are the

within-segment score cut-offs of Table 6 (0.523–0.545); flags are multivariate, so a high value on one feature raises the chance of review but never decides it.

Table 14: Empirical single-feature flag thresholds (pooled): the feature value above which the flag rate reaches 10%, 25% and 50%, and the flag rate among the top 1% of each feature.

Feature	Value for flag rate $\geq 10\%$	$\geq 25\%$	$\geq 50\%$	Flag rate in top 1% (%)	Feature p99
Gross_rent_amt	6,833.6	—	—	18.2	58,116.8
Other_foreign_inc_amt	1.0	380.0	—	26.9	1,710
Tot_CY_CG_amt	2.0	—	—	16.2	69,780.8
Spr_Prsnl_Contr	548.0	—	—	17.4	59,355.8
Gift_amt	651.0	—	—	16.3	2,634.1
Net_CG_amt	3.0	—	—	16.9	28,761.3
Asbl_forgn_source_incm_amt	2.0	—	—	14.6	5,926.1
Unfranked_Div_amt	2.0	—	—	19.3	1,117.1
Rent_int_ded_amt	164.7	59,227.6	—	19.8	31,072.3
Spr_Emplr_Contr	19,260.2	—	—	16.1	30,573.3

Fig. 10 gives waterfall explanations for three flagged taxpayers. (a) The highest-ranked injected record, a wage earner into which inflated rental deductions were injected, was isolated mainly by items that were unusual before injection: personal superannuation contributions at 3.3 times income, dividend deductions of \$127,766, capital gains and a cost of managing tax affairs equal to income; the injected rental item (capital-works deductions of \$102,713) appears only ninth, which shows why the primary forests recover injected ratio anomalies poorly. (b) The highest-ranked natural profile is a landlord with foreign income, gifts exceeding income, a net capital gain of \$84,172 and deductions 3.3 times income, the lawful-but-rare profile that dominates the review list. (c) The borderline record, a wage earner exactly at the 2% cut-off, is isolated by a termination payment of \$28,116 and other income of \$101,728 alongside small dividend items, each contributing modestly. Each explanation is a statement about the model, that these reported amounts and ratios are what isolated the record from its peers, and directs an officer to the items worth checking.

Fig. 10 Local SHAP explanations for three flagged taxpayers (contribution towards ‘anomalous’, feature values shown in AUD or as ratios)

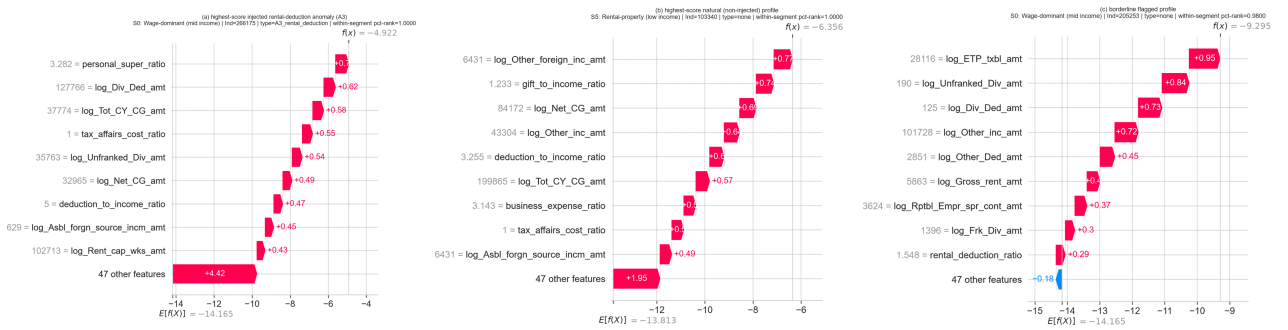


Figure 10: Local SHAP explanations for three flagged taxpayers (contribution towards “anomalous”).

For the Random Forest (Fig. 11) the taxable-income ratio dominated (mean $|\text{SHAP}|$ 0.095), followed by the deduction-to-income, rental-deduction and business-expense ratios (0.048, 0.035,

0.028), exactly the quantities the injection mechanisms perturbed. The rank correlation between the Random Forest and Isolation Forest importance orderings was -0.07 : the models attend to different features because they were built for different tasks.

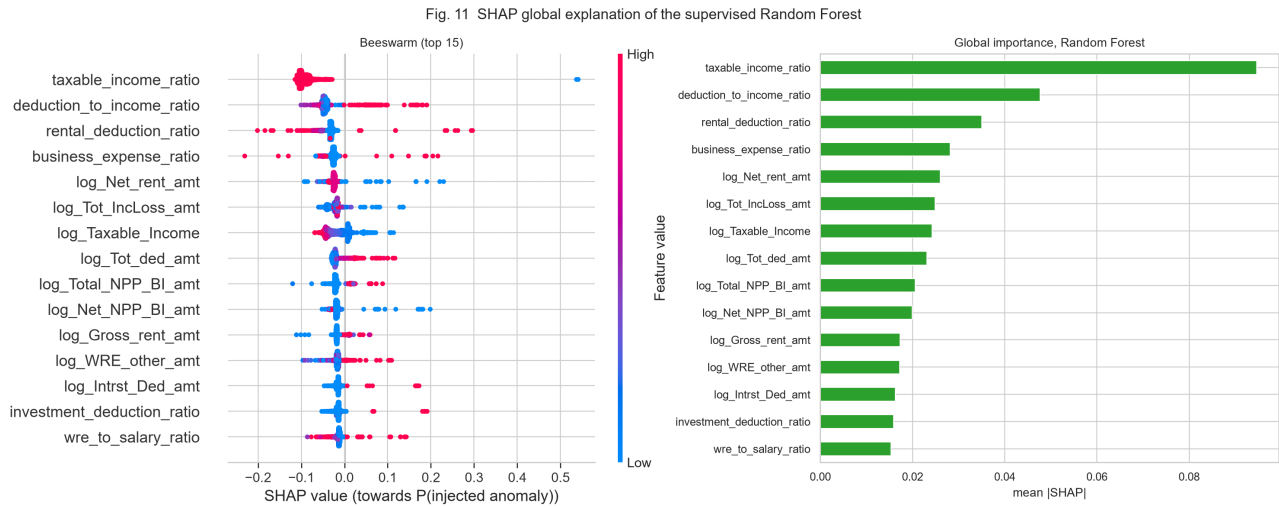


Figure 11: SHAP global explanation of the supervised Random Forest.

In a compliance context the SHAP output says that a high deduction-to-income relationship, large rental interest deductions and large investment-type income contributed strongly to the segment forests' classification of a taxpayer as unusual, and that an internally inconsistent taxable income was the single most informative signal for the supervised model. None of this establishes that any taxpayer acted improperly; large capital gains or foreign income are lawful and common at the top of the income distribution.

5 Discussion

5.1 Limitations

- **No genuine compliance labels were trained.** The sample file has no audit outcomes, hence every detection metric is measured against injected anomalies, and the supervised model’s near-perfect performance reflects the learnability of five documented mechanisms, not the detectability of real non-compliance. A statistical anomaly, a potential compliance-risk indicator and confirmed non-compliance are different things. In this study only statistical anomaly was measured.
- **False positives.** Ninety-four percent of flags were unmodified profiles, with large lawful capital gains or foreign income for example. Unsupervised flags are a starting point for review, not findings, and evaluating them against injected labels understates their usefulness.
- **Privacy, confidentiality and the sample.** The file is confidentialised and limited to a 2% sample whose extreme upper tail is compressed relative to the population. Full administrative data would face stricter governance and identify rarer profiles.
- **Single year (2022–23).** Unusual records in one year could be a normal transition such as retirement, a property sale or a business start-up. Multi year features are unavailable as taxpayer details are confidentialised.
- **Segment dependent results.** Anomaly scores are related only to the segment the taxpayer belong, so taxpayers change in segment could change the results.
- **Limitations of SHAP.** Shapley attributions depend on the feature set and on correlated features, can be unstable for near-tied contributions, explain the model rather than the data-generating process, and can be manipulated by adversarially constructed models [33, 34].

No flag produced here constitutes evidence of tax risk, fraud or evasion, and the framework is not described as detecting it either.

6 Conclusion and Recommendations

This paper proposed and implemented a four stage framework that segments taxpayers into peer groups, detects anomalous reporting within each group using Isolation Forest, tests the learnability of documented anomaly mechanisms with supervised classifiers, and explains segment membership and anomaly flags with SHAP. On the ATO 2022–23 sample file, segmentation on a nine-dimensional income-composition space produced eight well-separated, tax-interpretable peer groups with silhouette score 0.65, showing that the value of taxpayer segmentation depends on choosing a space that describes what kind of return is filed rather than how much is reported.

The segment based Isolation Forest changed more than half of a 2% review list relative to a population-wide forest and removed the systematic over-flagging of rare taxpayer types. The evaluation did not support the assumption that segmentation by itself makes the detector more sensitive: recall of injected ratio-type anomalies was no higher, income confounding within the large wage segment persisted, and the ablation showed that the feature space matters at least as much as the reference population. Isolation Forest remains an appropriate primary detector for this data, fast, scale-invariant and exactly explainable, but its flags are dominated by rarely reported items and it is weak on single-feature inconsistencies. Supervised models learned the injected mechanisms almost perfectly yet disagreed with the forest on almost every natural flag, clarifying the complementary roles of the paradigms: supervised models reproduce what is already known, unsupervised models surface what is not. SHAP made both models inspectable at the level of the return, with a surrogate bridging the clustering stage, and every explanation was verified against the model’s own score. For audit-risk assessment the message is that a segment-conditional, explainable anomaly score is a credible prioritisation tool, provided its output is treated as a signal for human review, its budget is set as a policy choice, and its sensitivity is measured against documented mechanisms rather than assumed.

6.1 Recommendations and Future Research

Future research should pursue hybrid designs in which unsupervised scores become features or weak labels for supervised models; deep autoencoders and graph-based detection linking taxpayers to agents, employers, partnerships and trusts [35]; temporal modelling across years so that transitions are distinguished from irregularities; anomaly detection in a scale-free behavioural feature space, which the ablation indicates is the more promising direction for peer-group designs; fairness evaluation and bias mitigation so that review burden does not fall disproportionately on particular occupations, regions or age groups [36]; and human-in-the-loop studies in which auditor feedback on explained flags is fed back into the models.

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Appendix A - Project Repository and Supporting Files

All project files, including the source code, notebooks, result tables and figures, are available in the project repository:

<https://github.com/jundram/Research>

Appendix B - Model Feature Set

Table 15 lists every dollar-amount variable in the ATO sample file with the share of taxpayers reporting a non-zero value and the screening decision described in Section 3.3. Table 16 gives summary statistics for the twelve behavioural features that were combined with the 44 retained amounts.

Table 15: Variable screening: share of taxpayers with a non-zero value (%) and screening decision for all 57 amount variables.

Variable	Non-zero (%)	Decision	Variable	Non-zero (%)	Decision
Sw_amt	79.65	kept	WRE_car_amt	23.47	kept
Alow_ben_amt	26.76	kept	WRE_trvl_amt	9.77	kept
ETP_txbl_amt	1.89	kept	WRE_uniform_amt	43.11	kept
Grs_int_amt	63.40	kept	WRE_self_amt	5.69	kept
Aust_govt_pnsn_allw_amt	10.34	kept	WRE_other_amt	54.89	kept
Unfranked_Div_amt	8.11	kept	Div_Ded_amt	2.38	kept
Frk_Div_amt	18.80	kept	Intrst_Ded_amt	1.08	kept
Dividends_franking_cr_amt	18.64	dropped: redundant	Gift_amt	27.69	kept
Net_rent_amt	14.10	kept	Non_emp_spr_amt	4.18	kept
Gross_rent_amt	14.03	kept	Cost_tax_affairs_amt	39.03	kept
Other_rent_ded_amt	14.02	dropped: redundant	Other_Ded_amt	5.80	kept
Rent_int_ded_amt	10.70	kept	Tot_ded_amt	76.10	kept
Rent_cap_wks_amt	7.47	kept	PP_loss_claimed	0.13	dropped: sparse
Net_farm_management_amt	0.12	dropped: sparse	NPP_loss_claimed	0.42	dropped: sparse
Net_PP_BI_amt	0.49	dropped: sparse	Rep_frng_ben_amt	6.63	kept
Net_NPP_BI_amt	10.17	kept	Asbl_forgn_source_incm_amt	7.49	kept
Total_PP_BI_amt	0.43	dropped: sparse	Net_fincl_invstmt_lss_amt	0.66	dropped: sparse
Total_NPP_BI_amt	9.92	kept	Rptbl_Empr_spr_cont_amt	9.47	kept
Total_PP_BE_amt	0.44	dropped: sparse	Cr_PAYG_ITI_amt	9.07	kept
Total_NPP_BE_amt	9.25	dropped: redundant	TFN_amts_wheld_gr_intst_amt	9.37	kept
Net_CG_amt	6.96	kept	TFN_amts_wheld_divs_amt	0.58	dropped: sparse
Tot_CY_CG_amt	9.11	kept	Hrs_to_prepare_BPI_cnt	0.63	dropped: sparse
Net_PT_PP_dsn	1.32	kept	Taxable_Income	97.69	kept
Net_PT_NPP_dsn	13.19	kept	Help_debt	14.99	kept
Taxed_othr_pnsn_amt	0.23	dropped: sparse	Spr_Emplr_Contr	76.55	kept
Untaxed_othr_pnsn_amt	1.27	kept	Spr_Prsnl_Contr	10.72	kept
Other_foreign_inc_amt	6.59	kept	Spr_Othr_Contr	3.63	kept
Other_inc_amt	8.85	kept	Spr_Ttl_Acnt_Bal	90.65	kept
Tot_IncLoss_amt	98.97	kept			

Table 16: Summary statistics of the twelve behavioural features ($n = 321,890$).

Feature	Mean	SD	Min	Median	P90	P99	Max
deduction_to_income_ratio	0.047	0.176	0.000	0.015	0.100	0.453	5.000
wre_to_salary_ratio	0.031	0.145	0.000	0.008	0.071	0.294	5.000
business_expense_ratio	0.070	0.482	-113.302	0.000	0.000	1.248	10.000
rental_deduction_ratio	0.165	0.551	0.000	0.000	0.688	2.193	10.000
taxable_income_ratio	0.999	0.021	0.400	1.000	1.000	1.000	1.000
investment_deduction_ratio	0.034	0.501	0.000	0.000	0.000	0.263	10.000
gift_to_income_ratio	0.003	0.034	0.000	0.000	0.003	0.043	2.000
tax_affairs_cost_ratio	0.005	0.044	0.000	0.000	0.006	0.085	1.000
personal_super_ratio	0.051	0.387	0.000	0.000	0.004	1.197	5.000
payg_instalment_ratio	0.015	0.107	0.000	0.000	0.000	0.289	2.000
items_reported	10.971	4.141	0.000	11.000	16.000	23.000	36.000
lodged_via_agent	0.629	0.483	0.000	1.000	1.000	1.000	1.000

Appendix C - Injected Anomaly Design and Audit

Table 17 records the eligible pool and the number of records modified for each of the five injected mechanisms of Section 3.3, and Table 18 shows the median deduction-to-income ratio and median taxable income of the modified records before and after injection, confirming that each mechanism changed the intended quantity.

Table 17: Injected anomaly mechanisms: eligible pool and number of records injected.

Type	Description	Eligible pool	Injected
A1_wre_inflation	Work-related expenses inflated to 25-45% of salary	218,028	644
A2_business_expense	Business expenses inflated to 150-300% of business income	23,526	644
A3_rental_deduction	Rental deductions inflated to 300-500% of gross rent	34,191	644
A4_taxable_identity	Taxable income reported at 40-70% of income less deductions and losses	235,253	644
A5_invest_ded_no_income	Investment deductions of 15-35% of income with zero investment income	69,700	643

Table 18: Injection audit: median deduction-to-income ratio and median taxable income (AUD) of injected records before and after modification.

Type	Ded./income before	Ded./income after	Taxable income before	Taxable income after
A1_wre_inflation	0.022	0.335	74,279.5	49,909.5
A2_business_expense	0.005	0.026	47,180.5	0.0
A3_rental_deduction	0.018	0.110	70,280.5	6,219.5
A4_taxable_identity	0.020	0.020	72,182.5	38,723.0
A5_invest_ded_no_income	0.021	0.308	64,319.0	45,284.0

Appendix D - Supplementary Results

This appendix contains supporting results referred to in Chapter 4 but not reported there in full: the clustering algorithms compared with K-Means (Table 19), the occupation mix of

each segment (Table 20), the five-fold cross-validation of the supervised models (Table 21), the SHAP validity check for each segment forest (Table 22), the global SHAP importance of the supervised Random Forest shown in Fig. 11 (Table 23), and the fifteen highest-scoring records of the segmented Isolation Forest with their top SHAP drivers (Table 24).

Table 19: Alternative clustering algorithms compared with K-Means on the segmentation features.

Model	Silhouette	Davies–Bouldin	Smallest segment (%)
K-Means (k=8)	0.653	0.719	1.31
Gaussian Mixture (diag, k=8)	0.149	2.585	4.20
Gaussian Mixture (full, k=8)	0.202	2.643	4.19
Ward agglomerative (k=8, 8,000-row sample)	0.642	0.711	0.94

Table 20: Occupation mix of each segment (% of taxpayers in the segment).

Segment	Clerical/admin	Community/personal service	Labourers	Machinery operators	Managers	Not stated	Other	Professionals	Sales	Technicians/trades
S0: Wage-dominant (mid income)	12.5	12.5	10.1	6.2	12.8	0.1	3.8	22.8	8.2	11.0
S1: Pension/retirement (low income)	4.5	7.5	8.5	2.4	1.8	62.1	2.1	3.6	5.0	2.4
S2: Business-dominant (mid income)	2.3	4.5	4.0	1.9	2.6	70.3	0.9	6.8	1.8	4.8
S3: Investment-income (low income)	3.2	1.4	0.7	0.6	4.6	82.7	0.7	4.2	0.9	1.0
S4: Other-income (mid income)	4.4	5.4	3.8	2.5	5.4	59.0	3.1	10.6	2.5	3.4
S5: Rental-property (low income)	5.3	2.9	1.7	0.9	5.8	71.0	1.0	7.2	2.0	2.2
S6: Trust/partnership (mid income)	7.2	3.0	1.7	0.7	7.0	67.3	1.2	6.6	3.4	2.0
S7: No/minimal reported income	0.4	0.5	0.6	0.3	0.4	96.3	0.4	0.4	0.3	0.4

Table 21: Five-fold stratified cross-validation of the supervised classifiers (mean $\pm$ SD).

Model	F1	ROC-AUC	PR-AUC	Training time (s)
Random Forest	0.920 $\pm$ 0.009	0.9997 $\pm$ 0.0000	0.970 $\pm$ 0.005	534.7
Hist. Gradient Boosting	0.856 $\pm$ 0.028	0.9980 $\pm$ 0.0010	0.974 $\pm$ 0.004	38.5

Table 22: SHAP validity check: Spearman correlation between the sum of SHAP values and the Isolation Forest score in each segment.

Segment	Records explained	Spearman (SHAP sum vs. score)
S0: Wage-dominant (mid income)	600	1.000
S1: Pension/retirement (low income)	600	1.000
S2: Business-dominant (mid income)	600	1.000
S3: Investment-income (low income)	600	1.000
S4: Other-income (mid income)	409	1.000
S5: Rental-property (low income)	486	1.000
S6: Trust/partnership (mid income)	589	1.000
S7: No/minimal reported income	385	1.000

Table 23: Global SHAP importance of the supervised Random Forest (top 15 of 56 features).

Rank	Feature	Mean SHAP
1	taxable_income_ratio	0.0947
2	deduction_to_income_ratio	0.0477
3	rental_deduction_ratio	0.0350
4	business_expense_ratio	0.0282
5	log_Net_rent_amt	0.0260
6	log_Tot_IncLoss_amt	0.0249
7	log_Taxable_Income	0.0242
8	log_Tot_ded_amt	0.0231
9	log_Total_NPP_BI_amt	0.0206
10	log_Net_NPP_BI_amt	0.0199
11	log_Gross_rent_amt	0.0173
12	log_WRE_other_amt	0.0172
13	log_Intrst_Ded_amt	0.0163
14	investment_deduction_ratio	0.0159
15	wre_to_salary_ratio	0.0153

Table 24: Fifteen highest-scoring records of the segmented Isolation Forest with their three largest SHAP contributions (values in AUD unless a ratio).

Record	Seg.	Injected	Score	Total income	Deductions	Top SHAP drivers
87459	S0	–	0.7364	245,409	45,445	log_Div_Ded_amt=3,550.00; log_Spr_Othr_Contr=43,530.00; log_Tot_CY_CG_amt=12,529.00
56577	S6	–	0.6753	1,490,569	15,289	log_Unfranked_Div_amt=2,991.00; log_WRE_uniform_amt=487.00; log_WRE_car_amt=3,580.00
85576	S2	A2_business_expense	0.7483	–427,401	117,241	log_Div_Ded_amt=51,135.00; log_Other_foreign_inc_amt=12,200.00; gift_to_income_ratio=1.09
259367	S1	–	0.7139	142,185	59,604	log_Intrst_Ded_amt=15,271.00; log_Tot_CY_CG_amt=28,467.00; log_Gross_rent_amt=36,687.00
84977	S3	–	0.6684	–2,210	26,806	deduction_to_income_ratio=5.00; log_Untaxed_othr_pnsn_amt=58,998.00; personal_super_ratio=5.00
220906	S7	–	0.7140	19	9,302	deduction_to_income_ratio=5.00; wre_to_salary_ratio=5.00; log_TFN_amts_wheld_gr_intst_amt=6.00
2684	S4	–	0.6663	1,327,612	512,041	log_Intrst_Ded_amt=328,522.00; log_Div_Ded_amt=22,516.00; business_expense_ratio=10.00
103340	S5	–	0.6505	–70,235	3,255	log_Other_foreign_inc_amt=6,431.00; gift_to_income_ratio=1.23; log_Net_CG_amt=84,172.00
266175	S0	A3_rental_deduction	0.7168	–410,927	132,117	personal_super_ratio=3.28; log_Div_Ded_amt=127,766.00; log_Tot_CY_CG_amt=37,774.00
284942	S0	–	0.7158	559,550	61,874	log_Tot_CY_CG_amt=20,252.00; log_Div_Ded_amt=51,135.00; log_Net_CG_amt=15,840.00
84356	S0	–	0.7077	1,186,484	149,708	log_Tot_CY_CG_amt=73,500.00; log_Unfranked_Div_amt=18,787.00; log_Asbl_forgn_source_incmt_amt=6,008.00
190823	S0	–	0.7070	144,621	94,550	log_Tot_CY_CG_amt=56,774.00; log_Div_Ded_amt=29,306.00; log_Unfranked_Div_amt=10,498.00
25520	S0	–	0.7063	199,428	25,500	log_Tot_CY_CG_amt=28,890.00; log_Asbl_forgn_source_incmt_amt=226,914.00; log_Spr_Othr_Contr=6,140.00
2694	S0	–	0.7032	608,399	50,805	log_Div_Ded_amt=32,715.00; log_Tot_CY_CG_amt=15,116.00; log_Spr_Prsnl_Contr=14,564.00
302688	S0	A3_rental_deduction	0.7003	–87,400	9,507	deduction_to_income_ratio=5.00; personal_super_ratio=2.18; tax_affairs_cost_ratio=0.32